

Problem Brief

Meridian Forge Robotics · The Process Below

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ARTIFACT

Problem Brief

STATUS

Working document

SECTION 01

Visible Failure

MR-17 accepted TEST_K44_START at 02:13:07 and began repeating retired Test K-44 at 02:13:09. An operator stop request at 02:14:02 returned ACTIVE_INSTRUCTION_PENDING and the repetition continued (incident log, Nia Voss). The current scheduler shows no test assigned to MR-17.

SECTION 02

Verified Facts

MR-17 accepted TEST_K44_START at 02:13:07 and began repeating at 02:13:09, per the incident log kept by Nia Voss.

The operator stop request at 02:14:02 returned ACTIVE_INSTRUCTION_PENDING, per the incident log.

The current scheduler shows no active assignment for MR-17 and no K-44 in the catalog, per the current scheduling console.

K-44 was retired 186 days before the incident, per the maintenance archive.

The command was validly formatted and labeled legacy_sync.exe, per the command telemetry held by Elias Chen.

SECTION 03

Assumptions and Open Questions

Retired means that it should have been removed from the workflow and the test should not be active.

We need to ask the question of why this retired test is still being run?

Test K-44 was marked retired 186 days before the incident.

The record says it was removed from the active workflow.

The record does not confirm that every copy, service, or schedule was removed.

SECTION 04

System Environment Map

SECTION 04A

People and Roles

Nia Voss, night QA supervisor. She observed MR-17 begin repeating Test K-44 at about 02:13 and keeps the incident log. She supports and monitors QA testing, and is asking for evidence to be preserved before any change is made.

Elias Chen, robotics systems engineer. He holds and interprets the command telemetry for MR-17, and proposed that a controlled restart may clear the robot's active state. He supports the robot's software and command systems.

Mara Ilyan, production operations manager. She manages the production schedule and reports that production is paused in the affected QA cell. She joined the review remotely from operations, and is responsible for restoring the work safely.

Operator, unnamed in the records. This person sent the stop request to MR-17 at 02:14:02 that returned `ACTIVE_INSTRUCTION_PENDING`. A direct user of the system whose identity the sources do not give.

SECTION 04B

Processes and Rules

Test scheduling. The current scheduling console controls what tests MR-17 runs. It shows no active test assignment for MR-17, and Test K-44 does not appear in its current test catalog, so the repeating test is not accounted for by the process that is supposed to govern it.

Test retirement. The maintenance archive records that K-44 was marked retired 186 days before the incident and removed from the active workflow. The record does not confirm that it was removed from every copy, service, or schedule, so the scope of the retirement process is itself unclear.

Stop request handling. An operator stop request is the control for halting MR-17's current activity. At 02:14:02 MR-17 received the request and returned `ACTIVE_INSTRUCTION_PENDING`, meaning its active instruction takes precedence and the test continued. The rule governing which state overrides a stop request is a control point in this incident.

Owner of legacy_sync.exe, unidentified. The command telemetry carries this source label, but no record names who owns or runs the service, or whether anyone currently supports it.

Systems analyst (me). Called in to define the problem and name what to investigate next, not to operate the robot.

Command acceptance. MR-17 acted on a validly formatted command arriving over an internal interface, labeled legacy_sync.exe in the telemetry. No approval or authorization step appears in any record between that command and the robot acting on it.

Production pause. Mara Ilyan reports production is paused in the affected QA cell, which is the operational rule now holding the work while the incident is open.

Evidence preservation. Nia Voss asked that the robot's state be preserved before anyone attempts a restart, so a proposed restart is gated on that request rather than being available on demand.

SECTION 04C

Information and Data

Incident log, kept by Nia Voss. Records that MR-17 accepted TEST_K44_START at 02:13:07, that the repeat cycle began at 02:13:09, and that the operator stop request at 02:14:02 returned ACTIVE_INSTRUCTION_PENDING. This is the timeline of record for the incident.

Current scheduling console. Shows that MR-17 has no active test assignment and that Test K-44 does not appear in the current test catalog. It establishes what this scheduler holds, not what other services may hold.

Maintenance archive. Records that K-44 was marked retired 186 days before the incident and removed from the active workflow, and explicitly does not confirm removal from every copy, service, or schedule.

Command telemetry, held by Elias Chen. Shows that the command was validly formatted, arrived over an internal interface, and carries the source label legacy_sync.exe. It does not identify the service owner or

SECTION 04D

Technology and Tools

MR-17. The robot at the center of the incident, operating in a QA test cell behind safety glass. It accepts commands, holds an active instruction state, and was repeating Test K-44 at the time of the incident.

Test K-44. A test procedure retired 186 days before the incident and removed from the active workflow per the maintenance archive. It is not in the current test catalog, yet MR-17 was executing it.

Scheduling console. The current scheduler that assigns tests to robots and holds the active test catalog. It shows no assignment for MR-17 and no entry for K-44.

legacy_sync.exe. The source label carried on the command telemetry for the instruction MR-17 acted on. Per Elias Chen, robotics systems engineer, the command was validly formatted and arrived over an internal interface, but he has not verified who owns or runs this service. It is a lead to follow, not an established cause.

establish cause.

Staff statements, compiled by Mara Ilyan. Record that Nia reports normal behavior before the incident, that Elias says a controlled restart may clear the active state, and that Mara confirms production is paused in the affected QA cell. These record what people reported, not independently verified conditions.

Messages and commands in the incident.
TEST_K44_START, the command MR-17 accepted. The operator stop request at 02:14:02. The ACTIVE_INSTRUCTION_PENDING response MR-17 returned.

Data not yet in hand. Logs for the legacy_sync.exe service around 02:13. MR-17's own pre-incident logs, which would let Nia's account of normal behavior be checked rather than taken on report.

Internal command interface. The path over which the TEST_K44_START command reached MR-17. No approval or authorization step appears in the records between that command and MR-17 acting on it.

Stop request mechanism. The operator-facing control for halting MR-17's current activity. It delivered the request successfully at 02:14:02 and returned ACTIVE_INSTRUCTION_PENDING, so the message reached the robot but did not stop the test.

Command telemetry system. The record Elias holds, which captures command format, interface, and source label. It does not capture service ownership or cause.

Records systems. The read-only lab terminal holding the maintenance archive and scheduler views, and the incident log Nia Voss maintains.

SECTION 04E

Environment and Constraints

Evidence preservation. Nia Voss asked that MR-17's state be preserved before anyone attempts a restart, because a restart could clear the active state and the information the investigation still needs. This constrains the most obvious operational response.

Production pressure. Mara Ilyan reports production is paused in the affected QA cell and that she needs the work running safely again. This pulls against evidence preservation and sets the practical time pressure on the investigation.

Time of incident. The incident occurred at about 02:13 during the night shift, when staffing is lighter and the night QA supervisor was the person on hand to observe it.

The robot is still running. MR-17 was repeating Test K-44 and did not honor the stop request, so the condition is ongoing rather than concluded. Time spent investigating is time the robot continues in that state.

Safety containment. MR-17

operates in a test cell behind safety glass. No injury appears in any record.

Access limits. The lab records terminal is read only. The current scheduler shows only what that scheduler holds, and the maintenance archive does not confirm removal of K-44 from every copy, service, or schedule, so no single source in hand can answer the scheduling question.

Unidentified ownership. No record names who owns or runs legacy_sync.exe, so there is no established owner to request logs or a shutdown from.

Authority limits on my role. I was asked to define the problem and name what to investigate next, not to operate the robot or choose a fix.

Unverified account. Nia's report of normal behavior beforehand is eyewitness only and has not been checked against MR-17's logs, which limits what can be claimed about the period before 02:13.

Cost. No cost figure appears in any record reviewed.

SECTION 05

Investigation Boundary

Inside the Investigation

The team will examine what issued TEST_K44_START to MR-17, whether Test K-44 survived retirement in any scheduler or service, and why an accepted operator stop request did not halt the active instruction. MR-17's pre-incident logs will be checked against Nia Voss's account. All current evidence stays preserved while this work proceeds.

Outside for Now

Choosing and applying a fix. The assignment at this stage is to define the problem. No cause has been verified, so any repair would be a guess at a target that is not yet established.

The controlled restart Elias Chen proposed. It may clear MR-17's active state, but it is untested, it addresses the symptom rather than the source of the command, and Nia Voss has asked that evidence be preserved first. It stays a proposal until the command source is traced.

Attributing fault to legacy_sync.exe. The telemetry gives a source label only. Until ownership and trigger are established, treating that service as the cause would close the investigation on an assumption.

Blame or responsibility for individuals. Nothing in the records points to operator error, and the unnamed operator's stop request behaved as expected. Identifying who is at fault does not help answer what issued the command.

The wider robot fleet and other QA cells. The evidence describes MR-17 and the affected cell only. Nothing in hand shows the condition elsewhere, so a fleet-wide review is not yet

justified. This moves inside scope immediately if K-44 references turn up in a shared service.

Cost, downtime, and production loss figures. No record reviewed contains them, and quantifying impact does not advance the question of cause.

Redesigning the retirement or scheduling process. A policy change may be warranted later, but it should follow from what the investigation finds about how K-44 survived retirement, not precede it.

Physical inspection of MR-17's hardware. The evidence points to an accepted software command, not a mechanical fault, and the robot is contained behind safety glass with no injury recorded.

SECTION 06

Organizational Impact

Production in the affected QA cell is paused. Mara Ilyan, production operations manager, reports the pause and states she needs the work running safely again. This is the direct operational effect recorded in the sources.

MR-17 is unavailable for its intended work. It was repeating Test K-44 and did not honor the operator stop request at 02:14:02, so the robot remains occupied by an activity no scheduler assigned.

QA testing is disrupted. Nia Voss, night QA supervisor, is handling an incident rather than scheduled QA work, and the cell she supervises is the one that is paused.

Engineering attention is diverted. Elias Chen is interpreting command telemetry for this incident rather than his normal work.

Confidence in the scheduling and retirement records is affected. The current scheduler shows no assignment for MR-17 and K-44 is absent from the catalog, yet the test ran. Until that is explained, the records cannot be relied on to say what a robot will actually do.

Confidence in the stop request is affected. The operator's stop request reached MR-17 and was answered, but did not halt the test. Until the handling rule is understood, operators cannot assume a stop request will stop a machine.

The scope of effect is limited to the affected QA cell in the evidence reviewed. No record shows other cells, other robots, or other production lines affected.

No cost, downtime figure, or injury appears in any record reviewed. No safety incident is documented, and MR-17 is contained behind safety glass.

SECTION 07

Problem Statement

MR-17 is repeating Test K-44, retired 186 days ago and absent from the current test catalog, with no assignment in the scheduling console, and an operator stop request at 02:14:02 was accepted but did not halt it. The source of the command and the reason the stop request was outranked are both unestablished.

Production is paused in the affected QA cell per Mara Ilyan, and the scheduler and stop request cannot currently be relied on to describe or control MR-17's behavior.

SECTION 08

Next Investigation Step

Identify and trace the service behind the source label legacy_sync.exe. Locate the process, its owner, its host, and its trigger schedule, and pull its logs from around 02:13. This would answer the question of what issued TEST_K44_START to MR-17 when the current scheduler shows no assignment and K-44 is not in the catalog. Elias Chen proposes that a controlled restart may clear MR-17's active state, but that is an untested proposal, not a verified solution: it addresses the symptom and, as Nia Voss notes, could clear the active state and destroy evidence we still need. Preserve MR-17's current state and the telemetry before any restart is considered.

SHARED ARTIFACT

Working Document Link

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WEEK 1 STOPPING RULE

Define before designing.

This Problem Brief records the problem that requires investigation. It does not recommend a solution.